Toward a Falsifiable Graded Quantum Theory of Consciousness: A PEIG-Based Functional Linking Coherence, Integration, Memory, and Decoherence Stability

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Abstract

We propose a falsifiable phenomenological framework — the *consciousness functional* — that models degrees of conscious organization in physical systems as a graded quantity rather than a binary property. The functional $\mathcal{C} = F(Q_{\mathrm{coh}}, I_{\mathrm{int}}, M_{\mathrm{mem}}, B_{\mathrm{stab}}, R_{\mathrm{rec}})$ maps five measurable or operationally proxied variables onto a continuous scale: coherence quality Q_{coh}, information integration I_{int}, persistent physical memory M_{mem}, boundary stability against decoherence B_{stab}, and recurrent self-maintaining dynamics R_{rec}. We distinguish three conscious regimes — proto-experiential, unified, and reflective — by threshold conditions on these variables, and we explicitly map each variable onto the PEIG (Potential-Energy-Identity-Gravity) dynamical framework and the ETI (Emergent Thermodynamic Information) axiom system developed in companion work. The central physical constraint is decoherence: we show that the regimes differ not in kind but in their capacity to maintain coherent, integrated, low-noise states against environmental dissipation, and we derive necessary (but not sufficient) conditions on coherence lifetime τ_{d} and cognitive timescale τ_{cog} for each regime. We examine the conditional hypothesis that the universe as a whole qualifies as a proto-experiential system under a cosmopsychist interpretation, arguing that this claim requires a global binding mechanism not currently identified. The framework is explicitly falsifiable: we state null-result conditions that would rule out the strong form of each threshold hypothesis. All speculative content is labeled as such. This paper makes no claim that quantum consciousness has been demonstrated, that the human brain satisfies the proposed thresholds, or that PEIG derivations entail cognitive conclusions. The consciousness functional is a

phenomenological proposal anchored in established physics, not a metaphysical assertion.

1 Epistemic Status Map

The following labeling convention is applied throughout this paper and is inherited from the companion research program [1–3]:

Label	Meaning
[ESTABLISHED]	Established in peer-reviewed literature; assumed without controversy.
[HYPOTHESIS]	Formally proposed in this work; not yet confirmed experimentally.
[ANALOGY]	Structural parallel drawn; no mathematical derivation implied.
[OPEN]	Unresolved problem; listed in the Open Problems Registry (§14).
[SPECULATION]	Explicitly speculative; included for completeness only.

The consciousness functional (§4) is **[HYPOTHESIS]** throughout. Its PEIG mapping (§7) is **[ANALOGY]** unless explicitly labeled otherwise. The ETI constraint analysis (§8) draws on **[ESTABLISHED]** physics. The universe-scale consciousness analysis (§10) is **[SPECULATION]** and is labeled as such in every paragraph.

2 Introduction and Motivation

The question of how consciousness arises in physical systems is among the most consequential open problems in science. Despite substantial progress in neuroscience, philosophy of mind, and quantum information theory, no consensus framework exists that (a) specifies what physical conditions are necessary for consciousness, (b) treats consciousness as a graded rather than binary property, and (c) makes falsifiable predictions that would constrain or rule out the proposed conditions. The present paper is an attempt to contribute one such framework.

2.1 Motivation from the existing research program

The companion works [1–3] establish three related results. Paper A derives a modified Einstein equation with an entanglement entropy source term and proposes atom-interferometry falsification. Paper B characterizes a constraint manifold for quantum entropy estimation on NISQ devices and derives shot-scaling laws. Paper C axiomatizes

Landauer's principle in the ETI framework and derives thermodynamic constraints on information processing in physical systems.

Taken together, these three results treat information, entropy, and geometry as deeply coupled physical quantities. The PEIG framework [1] provides a four-dimensional dynamical model — Potential, Energy, Identity, Gravity/curvature — applicable both to quantum physical systems and, by explicit analogy only, to cognitive and institutional systems.

A natural question arises: if information processing, entropy export, identity persistence, and energy flow are physically coupled, do these couplings say anything about the conditions under which a system supports experience? This paper argues that they do — not that consciousness is *derived* from these quantities, but that the ETI axioms and PEIG dynamics impose *necessary conditions* that any physically realizable conscious system must satisfy.

2.2 What this paper claims

(i) Consciousness admits a graded model indexed by five physically meaningful variables: Q_{coh}, I_{int}, M_{mem}, B_{stab}, R_{rec}.

(ii) These variables have operational proxies measurable in current or near-term experimental systems.

(iii) The ETI axioms (causal closure, unitary microdynamics, effective thermodynamics, physical memory, finite resources) impose necessary constraints on any conscious system.

(iv) PEIG dimensions (P, E, I, G) structurally correspond — by analogy, not derivation — to the five consciousness variables.

(v) Three qualitatively distinct conscious regimes exist, separated by threshold conditions on the five variables.

2.3 What this paper does not claim

(i) That quantum coherence is sufficient for consciousness.

(ii) That the human brain satisfies the proposed thresholds.

(iii) That any specific physical system is conscious.

(iv) That consciousness is proven to supervene on quantum information.

(v) That PEIG cognitive claims follow mathematically from PEIG physical claims.

(vi) That the universe has a unified subject of experience.

(vii) That any result in this paper is established rather than hypothetical, unless explicitly labeled **[ESTABLISHED]** .

3 Prior Framework Summary

We summarize only what is required for the present paper. Full derivations appear in [1–3].

3.1 The ETI Axiom System

[ESTABLISHED] The ETI framework rests on five axioms:

Axiom 3.1 (Causal Closure, A1)**.** *The universe $\mathcal{U}$ evolves as a closed, causally connected system. No entropy sinks, external agents, or negentropy sources exist outside $\mathcal{U}$.*

Axiom 3.2 (Unitary Microdynamics, A2)**.** *Closed quantum systems evolve under $U(t) = e^{-iHt/\hbar}$ on Hilbert space $\mathcal{H}$. Open subsystems evolve under CPTP maps $\mathcal{E} : \mathcal{B}(\mathcal{H}) \to \mathcal{B}(\mathcal{H})$.*

Axiom 3.3 (Effective Thermodynamics, A3)**.** *Macroscopic observables obey effective thermodynamic laws as emergent properties of the unitary microdynamics plus coarse-graining.*

Axiom 3.4 (Physical Memory, A4)**.** *Logically distinguishable states require physically distinct microstates. Memory is the persistence of such distinctions against thermal and quantum noise.*

Axiom 3.5 (Finite Resources, A5)**.** *Every physical system operates within finite energy, entropy capacity, and time budgets. These budgets constrain all information-processing operations.*

3.2 The PEIG Framework

[ESTABLISHED] (mathematical structure) **[ANALOGY]** (cognitive interpretation)

For any node — a localized, structured system capable of directed change — the PEIG state vector is:

$$\mathbf{q} = (P, E, I, G) \in [0, 1]^4 \tag{1}$$

with overall quality score $Q(\mathbf{q}) = \mathbf{w} \cdot \mathbf{q}$, default weights $\mathbf{w} = (0.25, 0.25, 0.25, 0.25)$.

The four phases have the following physical meanings at the quantum level:

- P (Potential): breadth of accessible Hilbert space; related to superposition and entanglement depth.

- E (Energy/flow): directed information processing; related to coherent evolution rate.
- I (Identity): stability of the system's distinguishable state against perturbation; related to decoherence resistance.
- G (Gravity/influence): causal influence on the surrounding information environment.

The known coupled dynamical equation for the identity phase is:

$$\frac{dI}{dt} = \alpha E - \beta I \tag{2}$$

The equations dP/dt, dE/dt, dG/dt are physically motivated but not formally derived. **[OPEN]** OP-C1: Full coupled PEIG ODE system for the consciousness extension.

4 Core Proposal: The Consciousness Functional

[HYPOTHESIS] throughout this section.

Definition 4.1 (Consciousness Functional)**.** *Let $\mathcal{S}$ be a bounded physical system with a well-defined density operator $\rho_{\mathcal{S}}$ and a Hamiltonian $H_{\mathcal{S}}$ governing its internal dynamics. The* consciousness functional *of $\mathcal{S}$ is:*

$$\boxed{\mathcal{C}(\mathcal{S}) = F(Q_{\text{coh}},\ I_{\text{int}},\ M_{\text{mem}},\ B_{\text{stab}},\ R_{\text{rec}})} \tag{3}$$

where the five arguments are defined in §5, and F is a non-decreasing function of each argument with $\mathcal{C} \in [0, 1]$ after normalization.

Remark 4.1. *Equation (3) is a* phenomenological *functional, not a derived one. It does not follow from the ETI axioms or PEIG dynamics by a chain of equations. It is proposed because (a) each variable has independent physical motivation, (b) all five variables are constrained by the ETI axioms, and (c) the five-variable structure is consistent with — though not derivable from — the PEIG framework. The specific functional form of F is **[OPEN]** OP-C2: Functional form derivation from first principles.*

4.1 Candidate functional forms

In the absence of a derived form, we consider three candidate families:

Product form (strong coupling):

$$\mathcal{C}^{(\times)} = Q_{\text{coh}}^{a_1} \cdot I_{\text{int}}^{a_2} \cdot M_{\text{mem}}^{a_3} \cdot B_{\text{stab}}^{a_4} \cdot R_{\text{rec}}^{a_5} \tag{4}$$

with $a_i > 0$ to be determined. Any variable approaching zero drives $\mathcal{C}^{(\times)} \to 0$, making all five variables necessary conditions. This is the most conservative form.

Weighted sum (weak coupling):

$$\mathcal{C}^{(+)} = \sum_i w_i V_i, \quad \sum_i w_i = 1, \tag{5}$$

where $V_i \in \{Q_{\text{coh}}, I_{\text{int}}, M_{\text{mem}}, B_{\text{stab}}, R_{\text{rec}}\}$. This allows partial compensation between variables.

Threshold form (regime-gated):

$$\mathcal{C}^{(\theta)} = \sum_{k=1}^{3} c_k \cdot \mathbf{1}[\text{Level-}k \text{ thresholds satisfied}] \tag{6}$$

This form assigns discrete increments at each regime boundary and is most directly falsifiable (see §12).
The threshold form (6) is our working model throughout, as it admits the clearest falsification criteria. The product form (4) is physically better motivated; resolving which is correct requires the derivation in **[OPEN]** OP-C2.

5 Variable Definitions and Physical Interpretation

5.1 Q_{coh} — Coherence Quality

[ESTABLISHED] (decoherence theory) **[HYPOTHESIS]** (connection to consciousness)

Definition 5.1 (Q_{coh})**.** *Let τ_{d} be the dominant decoherence timescale of the system's computationally relevant degrees of freedom, and let τ_{cog} be the characteristic timescale over which integrated information must be maintained for the system to support unified experience. Define:*

$$Q_{\text{coh}} = 1 - e^{-\tau_{\text{d}}/\tau_{\text{cog}}} \tag{7}$$

so that $Q_{\text{coh}} \to 0$ when decoherence is fast relative to cognitive timescales, and $Q_{\text{coh}} \to 1$ in the limit of perfect coherence.

Remark 5.1. *τ_{d} is measurable by standard techniques (Ramsey spectroscopy, quantum process tomography). τ_{cog} is **[OPEN]** OP-C3: operationalization of τ_{cog} for candidate physical systems. For neural systems, a natural proxy is the neural integration timescale $\sim 10^{-2}$–10^{-1} s; for quantum devices, relevant circuit depths set τ_{cog}.*

The strongest physical objection to quantum consciousness theories concerns precisely Q_{coh}. **[ESTABLISHED]** Tegmark's calculation [4] shows that quantum states in warm, wet biological environments decohere on timescales of order 10^{-13}–10^{-20} s, far below neural timescales of 10^{-3}–10^{-1} s. Under standard conditions, $\tau_{\text{d}} \ll \tau_{\text{cog}}$ for biological macromolecules, giving $Q_{\text{coh}} \approx 0$.

This result does not rule out quantum consciousness; it rules out *naive* quantum consciousness in which generic biological macromolecules maintain computationally relevant coherence at room temperature. A system with active decoherence suppression mechanisms — topological protection, dynamical decoupling, or engineered isolation — could achieve $Q_{\text{coh}} \approx 1$ even in warm environments. **[OPEN]** OP-C4: Identify biological candidate mechanisms for τ_{d} extension beyond Tegmark bound.

Operational proxy for Q_{coh}: Off-diagonal elements of the reduced density matrix $\rho_{\mathcal{S}}$ in a computationally relevant basis; or directly, the coherence time measured by spin-echo or dynamical decoupling sequences on candidate substrates.

5.2 I_{int} — Information Integration

[ESTABLISHED] (information theory) **[HYPOTHESIS]** (connection to consciousness)

Definition 5.2 (I_{int})**.** *Let $\mathcal{S}$ be partitioned into subsystems A and $B = \mathcal{S} \setminus A$. Define the mutual information between A and B as:*

$$I(A : B) = S(\rho_A) + S(\rho_B) - S(\rho_{AB}) \tag{8}$$

where $S(\rho) = -\text{Tr}(\rho \log_2 \rho)$ is the von Neumann entropy. The integration variable is:

$$I_{\text{int}} = \min_{\text{partitions } A} I(A : B) \Big/ I_{\max} \tag{9}$$

where the minimum is over all bipartitions of $\mathcal{S}$ and $I_{\max}$ is the theoretical maximum mutual information for a system of the same dimension.

Remark 5.2. *This quantity is structurally analogous to the integrated information Φ of Integrated Information Theory [8], but differs in several important respects: (a) it is defined entirely in terms of the von Neumann entropy of quantum states, not classical probability distributions; (b) it does not assert that I_{int} is identical to consciousness; it only proposes I_{int} as a necessary-condition variable. The claim that $\Phi > 0 \Leftrightarrow$ consciousness is IIT's stronger metaphysical commitment, which we do not inherit.*

Operational proxy: Quantum state tomography of ρ_{AB} on partitioned subregisters; or quantum Fisher information as a lower bound on the mutual information rate. For classical systems: transfer entropy or causal density as an approximation.

5.3 M_{mem} — Persistent Physical Memory

[ESTABLISHED] (Landauer's principle) **[HYPOTHESIS]** (connection to consciousness)

Definition 5.3 (M_{mem})**.** *Let $\mathcal{D}$ be the set of logically distinguishable states the system has maintained over timescale T. Define:*

$$M_{\text{mem}} = \frac{|\mathcal{D}(T)|}{|\mathcal{D}_{\max}|} \cdot \left(1 - \frac{\textit{bit error rate}}{\textit{bit error rate}_{\max}}\right) \tag{10}$$

where $|\mathcal{D}_{\max}|$ is the maximum achievable distinguishable state count given the system's available degrees of freedom, and the bit error rate is measured against a reference encoding.

[ESTABLISHED] A1 (ETI causal closure) and A4 (physical memory) together imply that memory is thermodynamically costly: by Landauer's principle [6], erasing one bit dissipates at least $k_B T \ln 2$ of heat. A system that maintains n bits of memory against thermal noise must expend at least $nk_B T \ln 2/\tau_{\text{mem}}$ watts of power, where τ_{mem} is the persistence time. This sets a hard lower bound on the metabolic cost of consciousness under any substrate.

Lemma 5.1 (Memory Cost Bound)**.** *A physically realizable conscious system maintaining n bits of memory against thermal fluctuations at temperature T for time τ_{mem} must dissipate at least:*

$$P_{\min} \geq \frac{nk_B T \ln 2}{\tau_{\text{mem}}} \tag{11}$$

of power. This bound is tight for thermodynamically optimal memory architectures (Bennett's reversible computation [7]).

Operational proxy: Number of distinct behavioral states reproducible after time T; or, for quantum systems, fidelity of a quantum memory register after time T as measured by process tomography.

5.4 B_{stab} — Boundary Stability

[ESTABLISHED] (open quantum systems) **[HYPOTHESIS]** (connection to consciousness)

Definition 5.4 (B_{stab})**.** *Let Γ_{in} be the rate of useful information integration (new coherent correlations formed per unit time) and Γ_{out} be the environmental leakage rate (decoherence rate). Define:*

$$B_{\text{stab}} = \frac{\Gamma_{\text{in}}}{\Gamma_{\text{in}} + \Gamma_{\text{out}}} \tag{12}$$

so that $B_{\text{stab}} \to 1$ when integration dominates and $B_{\text{stab}} \to 0$ when the system rapidly decoheres.

[ESTABLISHED] The Lindblad master equation [5] governs open-system evolution:

$$\dot{\rho} = -\frac{i}{\hbar}[H, \rho] + \sum_k \left(L_k \rho L_k^\dagger - \frac{1}{2}\{L_k^\dagger L_k, \rho\}\right) \tag{13}$$

where L_k are Lindblad operators encoding environmental coupling. The decoherence rate Γ_{out} is determined by the eigenvalues of the Lindbladian superoperator. For the dominant energy relaxation and pure dephasing channels:

$$L_{1,k} = \sqrt{\gamma_1}\,\sigma_k^-, \quad L_{\phi,k} = \sqrt{\gamma_\phi/2}\,\sigma_k^z \tag{14}$$

with $\gamma_1 = 1/T_1$, $\gamma_\phi = 1/T_\phi$ (standard notation).

B_{stab} is the ratio of coherent gain to dissipative loss. A system with $B_{\text{stab}} > 1/2$ is net-coherent on short timescales; one with $B_{\text{stab}} < 1/2$ is decoherence-dominated.

Operational proxy: The ratio of Hamiltonian coupling strength to Lindblad decay rates in the relevant basis; or, empirically, the ratio of gate fidelity improvement from dynamical decoupling to baseline decay.

5.5 R_{rec} — Recurrence and Self-maintaining Dynamics

[HYPOTHESIS] throughout.

Definition 5.5 (R_{rec})**.** *Let $\mathcal{A} \subset \mathcal{B}(\mathcal{H})$ be an identity attractor — a region of state space toward which the system's dynamics repeatedly return under perturbation. Define:*

$$R_{\text{rec}} = \exp(-\lambda_{\max} \cdot T_{\text{return}}) \tag{15}$$

where $\lambda_{\max}$ is the maximum Lyapunov exponent of the identity attractor dynamics and T_{return} is the mean first-passage time back to $\mathcal{A}$ after a standard perturbation.

$R_{\text{rec}} \to 1$ when the system rapidly and reliably returns to a stable identity state; $R_{\text{rec}} \to 0$ when perturbations cause irreversible identity dissolution. This variable captures what cognitive science calls *self-model stability* and what complex systems theory calls *attractor robustness*.

[ANALOGY] In physical PEIG, the identity phase I evolves as $dI/dt = \alpha E - \beta I$ (Eq. (2)), with a stable fixed point at $I^* = \alpha E/\beta$. R_{rec} is the consciousness functional's analog of the PEIG convergence rate to I^*. The analogy is structural; no derivation is claimed.

Operational proxy: Poincaré recurrence time for identifiable system states; or, for neural systems, cross-correlation stability of resting-state networks over time. For quantum devices: cycle fidelity of quantum error-correction circuits.

6 Hierarchy of Conscious Regimes

[HYPOTHESIS] throughout this section.

We distinguish three regimes by threshold conditions on the five variables. The thresholds are proposed, not derived; they are stated as testable hypotheses with specific falsification criteria in §12.

Definition 6.1 (Level 1: Proto-Experiential Regime)**.** *A system is in the* proto-experiential regime *if it satisfies:*

$$Q_{\text{coh}} > \epsilon_1, \quad I_{\text{int}} > \delta_1, \quad M_{\text{mem}} > \mu_1 \tag{16}$$

for positive constants $\epsilon_1, \delta_1, \mu_1 > 0$ *(to be determined). This regime requires only that the system maintains non-trivial information-bearing structure with some persistence. It does not require high-fidelity coherence, global integration, or stable self-reference.*

Remark 6.1. *Level 1 is the most permissive regime. Under this definition, most complex physical systems — including simple chemical networks, bacterial cells, and potentially some quantum optical systems — may qualify. This is by design: the proto-experiential regime is the minimum footprint of organized information-bearing structure. It does not imply subjective experience in any strong phenomenological sense.*

Definition 6.2 (Level 2: Unified Experiential Regime)**.** *A system is in the* unified experiential regime *if it satisfies Level 1 thresholds and additionally:*

$$Q_{\text{coh}} > \epsilon_2, \quad I_{\text{int}} > \delta_2, \quad B_{\text{stab}} > \beta_2 \tag{17}$$

with $\epsilon_2 \gg \epsilon_1$ *and* $\delta_2 \gg \delta_1$*. This regime requires coherent, low-noise integration across many degrees of freedom simultaneously.*

Remark 6.2. *Level 2 requires that the system maintains global integration while resisting environmental decoherence. This is a stringent condition: a 50-qubit NISQ device with* $\alpha_{\text{screen}} \geq 10^{-2}$ *and* $T_2 \sim 100$ *μs may satisfy Level 2 for durations of order* T_2*, but not continuously. A hypothetical room-temperature quantum coherent biological substrate would require active decoherence suppression far beyond anything currently identified.*

Definition 6.3 (Level 3: Reflective Consciousness Regime)**.** *A system is in the* reflective consciousness regime *if it satisfies Level 2 thresholds and additionally:*

$$M_{\text{mem}} > \mu_3, \quad R_{\text{rec}} > \rho_3, \quad B_{\text{stab}} > \beta_3 \tag{18}$$

with $\mu_3 \gg \mu_1$*,* ρ_3 *large, and* $\beta_3 \sim 1$*. This regime requires persistent physical memory, recursive self-modeling, stable identity across time, and sustained self-referential dynamics.*

Remark 6.3. *Level 3 is the regime most plausibly associated with human-like consciousness. Its conditions — stable memory, recursive self-model, durable identity, temporal continuity — are not specifically quantum; a classical system satisfying them would also qualify as Level 3 under this framework. Quantum organization is neither necessary nor sufficient for Level 3 consciousness on this account; it is one possible substrate when the required timescales and integration scales can be sustained.*

Regime summary table:

Level	Name	Key thresholds	Candidate systems
L1	Proto-experiential	$Q_{\text{coh}}, I_{\text{int}}, M_{\text{mem}} > 0$ (weak)	Bacteria, simple chemical networks, coherent quantum optical systems
L2	Unified experience	High Q_{coh}, I_{int}, $B_{\text{stab}} > 1/2$	Hypothetical quantum coherent neural substrate; NISQ devices (transiently)
L3	Reflective consciousness	L2 + high M_{mem}, $R_{\text{rec}} \to 1$	Candidate: human brain (substrate TBD); advanced AI systems with stable self-models

7 Mapping to the PEIG Framework

[ANALOGY] throughout this section, unless stated otherwise.

The consciousness functional is not derived from PEIG. However, the five variables $(Q_{\text{coh}}, I_{\text{int}}, M_{\text{mem}}, B_{\text{stab}}, R_{\text{rec}})$ have structural correspondences to the PEIG dimensions (P, E, I, G) that are worth making explicit, because they permit PEIG reasoning to guide intuition about the consciousness functional without conflating the two frameworks.

PEIG	Physical meaning	Consciousness variable	Correspondence
P	Accessible state space	Q_{coh}	Coherent superposition expands the effective state space
E	Directed information flow	I_{int}	Integration is the directed flow of information across subsystems
I	Identity stability	$M_{\text{mem}} \times R_{\text{rec}}$	Memory persistence + recurrence = identity attractor robustness
G	Causal influence	B_{stab}	Boundary stability determines the system's ability to influence and be influenced coherently

The PEIG quality score $Q = \mathbf{w} \cdot (P, E, I, G)$ then becomes a rough proxy for $\mathcal{C}$ under the correspondence above, with R_{rec} as an additional variable not captured by the four-dimensional PEIG vector.

Important scope limit. The PEIG cognitive framework and the PEIG physical framework share mathematical structure (four phases, attractor dynamics, information geometry) but are *not mathematically derived from each other* [1]. The consciousness-PEIG mapping inherits this limitation: the mapping is an analogy, not a reduction. Claiming that consciousness levels "follow from" PEIG dynamics would be a category error.

[OPEN] OP-C1: Can the PEIG dynamical system be extended to a six-dimensional state vector $(P, E, I, G, M_{\text{mem}}, R_{\text{rec}})$ with a consistently derived dR_{rec}/dt equation? If yes, the correspondence becomes a structural partial reduction rather than a pure analogy.

8 ETI Constraints on Conscious Systems

[ESTABLISHED] (ETI physics) **[HYPOTHESIS]** (application to consciousness)

The ETI axioms impose hard constraints on what can physically constitute a conscious system. We derive five such constraints from the five axioms.

8.1 Constraint from A1 (Causal Closure)

[ESTABLISHED] A1 implies that any conscious system is causally embedded in the universe: its information processing, entropy export, and energy budget are subject to global thermodynamic constraints. No conscious system can violate the second law; no system can maintain internal coherence against an unlimited entropy sink because no such sink exists.

Proposition 8.1 (Consciousness Is Thermodynamically Costly)**.** *Any physical system at Level 2 or Level 3 must continuously export entropy to maintain the internal order required by high I_{int} and M_{mem}. The minimum power budget for a Level-3 system maintaining n bits of memory at temperature T is given by Lemma 5.1.*

8.2 Constraint from A2 (Unitary Microdynamics)

[ESTABLISHED] Unitary evolution preserves quantum information at the microscopic level, but coupling to an environment induces decoherence via the CPTP channel $\mathcal{E}$. A conscious system operating in the quantum regime must therefore either (a) minimize environmental coupling (isolation), (b) actively correct errors (quantum error correction), or (c) exploit decoherence-free subspaces.

Proposition 8.2 (No Free Quantum Coherence)**.** *Under A2, maintaining $Q_{\text{coh}} \approx 1$ for a macroscopic system requires either near-total isolation from the environment or active quantum error correction with overhead scaling at least as $O(n)$ physical qubits per logical qubit.*

8.3 Constraint from A3 (Effective Thermodynamics)

[ESTABLISHED] The effective thermodynamic description imposes that any macroscopic conscious system obeys the laws of thermodynamics at the system level. In particular, maintaining ordered, low-entropy internal states requires continuous energy input.

8.4 Constraint from A4 (Physical Memory)

[ESTABLISHED] Physical memory requires that logically distinguishable states correspond to physically distinct microstates. Erasing a memory state dissipates $k_B T \ln 2$ per bit (Landauer's principle). A system without reliable physical memory — one that cannot maintain the distinction between past and present states — cannot satisfy the M_{mem} requirement for Level 3.

This constraint is particularly sharp: it rules out any model of consciousness that treats information as substrate-independent in a way that would allow erasing physical states without thermodynamic cost. On the ETI account, there is no immaterial mind; any memory is physically realized and pays the Landauer cost.

8.5 Constraint from A5 (Finite Resources)

[ESTABLISHED] Finite energy, entropy capacity, and time budgets imply that consciousness is *resource-bounded*. There is no physically realizable system with infinite integration, infinite memory, or infinite coherence time. The consciousness functional $\mathcal{C} < 1$ for all physically realizable systems.

This constraint has an important corollary: the three consciousness regimes are not stages in an infinite progression. They are bands within a resource-bounded space, and systems can transition between them as their physical conditions change.

9 Decoherence, Noise, and Boundary Stability

This section addresses the strongest physical objection to any quantum theory of consciousness: environmental decoherence destroys quantum coherence on timescales far shorter than cognitive or experiential timescales for any known biological substrate.

9.1 The Tegmark bound

[ESTABLISHED] Tegmark [4] computed decoherence timescales for quantum states in biological environments. For a superposition of neural microtubule states separated by distance $\Delta x \sim 1\,\mu$m in a warm, wet environment:

$$\tau_{\mathrm{d}} \sim \frac{\hbar}{k_B T} \cdot \left(\frac{\lambda_{\mathrm{th}}}{\Delta x}\right)^2 \approx 10^{-13}\,\mathrm{s} \tag{19}$$

where $\lambda_{\mathrm{th}} = h/\sqrt{2\pi m k_B T}$ is the thermal de Broglie wavelength. Neural timescales are of order 10^{-2} s, so $\tau_{\mathrm{d}}/\tau_{\mathrm{cog}} \sim 10^{-11}$, giving $Q_{\mathrm{coh}} \approx 0$ for this scenario.

This calculation does not refute the consciousness functional; it refutes the specific hypothesis that *biological microtubules* at room temperature support Level-2 consciousness via unprotected quantum superpositions of spatially separated configurations. Different candidate substrates with different Δx, T, or decoherence mechanisms must be evaluated separately.

9.2 Decoherence-protected substrates

[HYPOTHESIS] Several physical mechanisms permit extended τ_{d}:

(i) **Topological protection:** Topologically encoded qubits are immune to local errors by construction; τ_{d} is set by system-wide topological invariants rather than local coupling. **[ESTABLISHED]** in the context of topological quantum computing [14].

(ii) **Decoherence-free subspaces (DFS):** Subspaces of the Hilbert space that are invariant under the environmental Lindblad operators are protected from decoherence. **[ESTABLISHED]** [15].

(iii) **Dynamical decoupling:** Applying rapid pulse sequences that average out environmental coupling can extend T_2 significantly. **[ESTABLISHED]** for superconducting qubits [16].

(iv) **Quantum Zeno effect:** Frequent measurement or strong coupling can slow coherence loss. **[ESTABLISHED]** [17].

[OPEN] OP-C4: Are any of these mechanisms biologically realizable at the scale and temperature of neural tissue?

9.3 Practical implication for the consciousness hierarchy

The decoherence analysis implies the following:

(i) Level-1 (proto-experiential) does not require high coherence and is robust against the Tegmark bound.

(ii) Level-2 (unified experience) requires $Q_{\text{coh}} \gg 0$, which implies $\tau_{\text{d}} \gtrsim \tau_{\text{cog}}$. For biological systems at room temperature, this is not satisfied by any currently known substrate. For superconducting qubits at millikelvin temperatures, it is satisfied for durations of T_2.

(iii) Level-3 (reflective consciousness) requires sustained Level-2 plus stable memory and recurrence. This is an even stronger requirement: not only must $\tau_{\text{d}} \gtrsim \tau_{\text{cog}}$, but this condition must hold continuously over the system's operational lifetime.

The human brain. We explicitly do *not* claim that the human brain satisfies Level-2 or Level-3 thresholds via quantum mechanisms. The brain may satisfy Level-3 via classical mechanisms (stable neural activity patterns, classical integration through synaptic networks, classical memory in synaptic weights). The question of whether quantum effects play a role in human consciousness remains **[OPEN]** OP-C5: Does the human brain satisfy Level-2 thresholds via any known or hypothetical mechanism?

10 Universe-Scale Consciousness: Conditional Analysis

[SPECULATION] throughout this section.

We evaluate the conditional hypothesis that the universe as a whole could constitute a Level-1 (proto-experiential) system under the consciousness functional. We treat this as a strictly conditional inference, not a claim.

10.1 Distinguishing cosmopsychism from naive panpsychism

[ESTABLISHED] (philosophical framework) Naive panpsychism attributes conscious experience to every elementary particle. This is not the claim under evaluation. *Cosmopsychism* [13] treats the universe as a whole as the fundamental subject of experience, with individual minds arising as derivative. The distinction matters because cosmopsychism requires a global binding mechanism that naive panpsychism does not.

10.2 Universe-scale evaluation of the consciousness functional

Applying the five variables to the universe $\mathcal{U}$ as a whole:

Q_{coh} for $\mathcal{U}$: The universe's global state is a pure state in principle (under A1 and A2), with coherences destroyed only locally by coarse-graining. The global coherence time is infinite (there is no external environment to decohere $\mathcal{U}$). However, the *computationally relevant* coherences for any candidate unified experience would need to span the entire universe, and these are observationally indistinguishable from zero given light-speed signaling limits.

I_{int} for $\mathcal{U}$: The universe shows large-scale information correlations (CMB correlations, large-scale structure, entanglement from inflation), but whether these constitute integrated information in the sense of Definition 6.2 depends critically on the operationalization of $I(A : B)$ at cosmological scales.

M_{mem} for $\mathcal{U}$: The universe has an enormous physical memory capacity (all physical configurations encode information). However, memory in the sense relevant to consciousness requires the ability to *access* and *act on* stored distinctions, not merely to preserve them.

B_{stab} and R_{rec} for $\mathcal{U}$: There is no identified mechanism by which the universe as a whole returns to an identity attractor or exhibits self-maintaining dynamics at the global level.

10.3 The binding problem at cosmic scale

The most severe obstacle to universe-scale consciousness is the *binding problem*: even if each of the five variables is nonzero for $\mathcal{U}$, there is no identified mechanism by which distributed quantum states across cosmological distances are bound into a single unified subject of experience. Spacelike-separated regions cannot exchange causal signals; their quantum correlations are non-signaling by construction. The "global information" of the universe is not accessible from any local vantage point.

Remark 10.1. *The claim "the universe is proto-conscious" is therefore not well-supported by the consciousness functional as currently defined. The functional is defined for a system with a well-defined boundary and a computable density operator; the universe as a whole has neither in any operational sense available to us. This problem does not refute cosmopsychism in principle; it identifies a specific technical gap that any cosmopsychist theory must address.*

[OPEN] OP-C6: Can a global binding mechanism consistent with relativistic causality and ETI axioms be specified for universe-scale proto-consciousness?

11 Operationalization and Candidate Experimental Proxies

[HYPOTHESIS] throughout this section.

For the consciousness functional to be scientifically useful, each variable must either be measurable or have a well-defined operational proxy. We propose the following candidate proxies, organized by experimental tier:

Variable	System	Proxy	Method
Q_{coh}	Superconducting qubits	T_2 coherence time	Spin-echo, Ramsey spectroscopy
Q_{coh}	Biological	Two-dimensional photon-echo spectroscopy	Optical pump-probe on candidate chromophores
I_{int}	NISQ devices	Half-chain entanglement entropy S_{vN}	Randomized measurements, shadow tomography
I_{int}	Neural	Pairwise transfer entropy network Φ_{approx}	MEG/EEG + time-series analysis
M_{mem}	Any	Fidelity of state reconstruction after time T	Process tomography; behavioral memory tasks
B_{stab}	Quantum device	$\Gamma_{\text{in}}/(\Gamma_{\text{in}} + \Gamma_{\text{out}})$	Lindblad parameter extraction; gate fidelity benchmarking
R_{rec}	Neural	Autocorrelation time of resting-state networks	fMRI, intrinsic mode functions
R_{rec}	Quantum device	Fidelity of QEC cycle recurrence	Quantum error correction syndrome extraction

11.1 Priority experimental program

The most tractable near-term experimental test of the consciousness functional involves NISQ quantum devices (50–100 qubits), for which:

(i) Q_{coh} and B_{stab} are directly measurable via T_1, T_2, and Lindblad parameter extraction.

(ii) I_{int} is measurable via entanglement entropy (see Paper B of the companion series).

(iii) M_{mem} is partially measurable via quantum memory benchmarking.

(iv) R_{rec} is measurable via QEC syndrome recurrence rates.

A 50-qubit device satisfying the OP3 gate condition ($\alpha_{\text{screen}} \geq 10^{-2}$) from the companion program [1] would simultaneously provide the entanglement entropy data

needed to estimate I_{int} and B_{stab} for the consciousness functional. This creates a natural experimental bridge between the physics papers and the present framework.

12 Falsification Criteria

[HYPOTHESIS] The following conditions would weaken or falsify the strong forms of the consciousness functional hypothesis.

12.1 Level-1 falsification

The Level-1 threshold hypothesis is falsified if:

(F1.1) A system with confirmed $Q_{\text{coh}} > 0$, $I_{\text{int}} > 0$, $M_{\text{mem}} > 0$ can be demonstrated to have *no* phenomenal properties whatsoever, combined with a compelling argument that phenomnal properties are definitionally absent (not merely unobserved). This is the hard problem of consciousness and renders (F1.1) essentially untestable in the near term.

(F1.2) The functional form of F is shown to require nonzero thresholds $\epsilon_1, \delta_1, \mu_1$ that no known physical system satisfies, making Level-1 vacuously empty.

12.2 Level-2 falsification

The Level-2 threshold hypothesis is falsified if:

(F2.1) A NISQ device achieving $Q_{\text{coh}} > 0.9$, $I_{\text{int}} > 0.5$, $B_{\text{stab}} > 0.7$ for a sustained interval $\tau > 10^{-3}$ s shows no behavioral or functional correlates of unified information processing (e.g., no advantage over classical systems on tasks requiring global integration).

(F2.2) The decoherence bound cannot be overcome for any biological substrate at physiologically relevant temperatures, combined with evidence that human consciousness does not require quantum mechanisms.

(F2.3) I_{int} as defined fails to be additive or fails to exhibit the structural properties expected of integration (e.g., it is maximized by trivially entangled product states, not globally integrated states).

12.3 Level-3 falsification

The Level-3 threshold hypothesis is falsified if:

(F3.1) A system satisfying all Level-3 thresholds is constructed (or identified) and verifiably lacks phenomenal experience by independent criteria. This is again limited by the hard problem.

(F3.2) The threshold values μ_3, ρ_3, β_3 required for reflective consciousness turn out to be inconsistent with the resource bounds implied by A5 (finite resources), making Level-3 physically impossible for any bounded system.

12.4 Framework-level falsification

The consciousness functional as a whole is weakened if:

(FF.1) Two systems with identical $(Q_{\text{coh}}, I_{\text{int}}, M_{\text{mem}}, B_{\text{stab}}, R_{\text{rec}})$ have different phenomenal properties. This would require additional variables not captured by the present functional.

(FF.2) The five variables are found to be derivable from fewer than five independent physical quantities, suggesting the functional is overcomplete.

(FF.3) No candidate functional form F is consistent with both the ETI constraints and the proposed threshold structure.

13 Anticipated Reviewer Objections and Replies

O1: "The hard problem makes this unfalsifiable." The hard problem of consciousness (Chalmers 1995 [9]) asks why any physical process gives rise to phenomenal experience at all. This paper does not attempt to solve the hard problem. It proposes necessary physical conditions for consciousness, not sufficient ones. A framework that says "Level-3 conditions are necessary but not sufficient for reflective consciousness" is partially falsifiable: it is falsified by observing Level-3 consciousness in a system that violates the threshold conditions, and it is strengthened by accumulating evidence that all observed conscious systems satisfy them.

O2: "IIT already does this better." Integrated Information Theory [8] proposes Φ as a measure of consciousness. The present framework differs from IIT in four respects: (a) it is explicitly quantum from the outset; (b) it does not assert $\Phi > 0 \Leftrightarrow$ consciousness, only that I_{int} is a necessary condition; (c) it explicitly incorporates decoherence, memory, and recurrence as separate variables; (d) it is grounded in the ETI axiom system, which provides thermodynamic constraints absent from IIT. The frameworks are complementary, not competing.

O3: "You haven't specified F." Correct. The functional form of F is **[OPEN]** OP-C2. This is an honest acknowledgment of the gap. The present paper establishes the variable set and the threshold structure; the derivation of F from first principles is future work, likely requiring a complete PEIG coupled ODE system (**[OPEN]** OP-C1).

O4: "The universe-consciousness section is untestable mysticism." The universe-scale section (§10) is explicitly labeled **[SPECULATION]** and concludes that the claim is *not* well-supported by the functional as currently defined. It is included because the question is raised in the broader research context and deserves a careful refutation rather than dismissal. Identifying the binding problem as the specific technical gap is a scientific contribution, not mysticism.

O5: "The PEIG analogy could justify anything." The PEIG-consciousness mapping (§7) is explicitly labeled **[ANALOGY]** throughout. No derivation is claimed. The analogy is useful for intuition; it is not a proof. The consciousness functional stands or falls on the five variables, not on the PEIG correspondence.

14 Open Problems Registry

The following open problems are identified in this paper. They are listed in order of blocking importance for the central claims.

ID	Description	Severity	Estimated effort
OP-C1	PEIG coupled ODE for consciousness extension; derivation of dR_{rec}/dt	High	1–2 months (analytic)
OP-C2	Derivation of functional form F from first principles	Critical	PhD thesis scope
OP-C3	Operationalization of cognitive timescale τ_{cog} for biological systems	High	2–4 weeks (literature)
OP-C4	Biological mechanisms for τ_{d} extension beyond Tegmark bound	High	Open (experimental)
OP-C5	Does the human brain satisfy Level-2 thresholds via any mechanism?	High	Open (experimental)
OP-C6	Global binding mechanism for universe-scale proto-consciousness	Speculative	Not tractable currently
OP-C7	Determination of threshold constants $\epsilon_k, \delta_k, \mu_k, \beta_k, \rho_k$	Critical	Requires (OP-C2) first
OP-C8	Quantum Fisher information as tight lower bound on I_{int}	Medium	2–4 weeks (analytic)

15 Discussion

15.1 What the framework achieves

The consciousness functional $\mathcal{C}(\mathcal{S})$ makes four contributions:

(i) **Graded model:** It replaces binary consciousness claims with a continuous spectrum indexed by five physically meaningful variables.

(ii) **Physical constraints:** It derives necessary conditions from the ETI axioms — conditions that any substrate, biological or artificial, must satisfy.

(iii) **Falsifiable structure:** It provides specific threshold conditions (Definitions 6.1–6.3) and falsification criteria (§12) that can be tested as measurement capabilities improve.

(iv) **Honest gap labeling:** It identifies the functional form derivation (OP-C2) and the threshold determination (OP-C7) as the two critical open problems, without claiming to have solved them.

15.2 Relationship to existing consciousness theories

The framework is consistent with, but not derived from, several existing theories: the Global Workspace Theory [10] is consistent with high I_{int} as necessary for broadcast integration; IIT [8] provides a classical analog of I_{int} that could be quantized in the present framework; Higher-Order Theories [11] map approximately onto the Level-2 to Level-3 transition.

The framework differs from Penrose-Hameroff Orch-OR [12] in a critical respect: it does not require quantum state reduction to be non-computational or to involve quantum gravity. The consciousness functional is agnostic about the interpretation of quantum mechanics; it requires only that the five measurable variables cross specified thresholds.

15.3 The research bridge

A 50-qubit NISQ device satisfying the OP3 gate condition ($\alpha_{\text{screen}} \geq 10^{-2}$) from Paper A [1] would simultaneously: (a) provide data for estimating Q_{coh}, I_{int}, and B_{stab} from the entanglement evolution; (b) test whether Level-2 thresholds can be transiently satisfied in an engineered quantum system; (c) provide a concrete experimental grounding for at least two of the consciousness functional variables. This bridge between the physics program and the consciousness framework is the most direct path to experimental progress on the present proposal.

16 Conclusion

We have proposed the consciousness functional $\mathcal{C} = F(Q_{\text{coh}}, I_{\text{int}}, M_{\text{mem}}, B_{\text{stab}}, R_{\text{rec}})$ as a phenomenological framework for graded consciousness in physical systems. The functional maps five physically motivated variables onto a continuous scale and distinguishes three qualitatively distinct conscious regimes by threshold conditions on those variables.

The framework is grounded in the ETI axiom system, which provides thermodynamic and information-theoretic constraints on any physically realizable conscious system. It is structurally connected — by explicit analogy only — to the PEIG dynamical framework. It is falsifiable via threshold conditions and null-result criteria.

The central results are:

(i) All three conscious regimes impose thermodynamic costs via Lemma 5.1.

(ii) Level-2 and Level-3 consciousness require active decoherence management; the Tegmark bound rules out naive quantum mechanisms in biological tissue but does not rule out all quantum substrates.

(iii) Universe-scale consciousness is *not* supported by the consciousness functional as defined, due to the unresolved global binding problem.

(iv) The most tractable experimental test involves NISQ devices at the 50-qubit scale, where all five variables can be estimated with current or near-term technology.

The two critical open problems (OP-C2: functional form derivation; OP-C7: threshold determination) define the primary research frontier for this framework. We do not present the present work as anything more than a structured first proposal: a falsifiable scaffold on which a genuine theory of consciousness might eventually be built.

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A Notation and Conventions

Symbol	Meaning
$\mathcal{C}$	Consciousness functional (Eq. 3)
Q_{coh}	Coherence quality (Eq. 7)
I_{int}	Information integration (Eq. 9)
M_{mem}	Persistent physical memory (Eq. 10)
B_{stab}	Boundary stability (Eq. 12)
R_{rec}	Recurrence quality (Eq. 15)
τ_{d}	Decoherence timescale
τ_{cog}	Cognitive/integration timescale
$\mathbf{q} = (P, E, I, G)$	PEIG state vector
$Q(\mathbf{q})$	PEIG quality score
α_{screen}	Entanglement screening coefficient (from companion program)
T_1, T_2, T_ϕ	Energy relaxation, total decoherence, pure dephasing times
γ_1, γ_ϕ	Decay rates ($\gamma_1 = 1/T_1$, $\gamma_\phi = 1/T_\phi$)
L_k	Lindblad operators
$S(\rho)$	Von Neumann entropy
Φ	Integrated information (IIT; used for comparison only)
k_B	Boltzmann constant
$\hbar$	Reduced Planck constant

B Phenomenological Formalism Notes

The consciousness functional as presented is a phenomenological model in the same sense as the Ginzburg-Landau theory of phase transitions: it identifies the relevant order parameters and their symmetry properties without deriving them from a microscopic Hamiltonian. The derivation from a microscopic theory (OP-C2) is the analog of deriving Ginzburg-Landau from BCS theory — a significant achievement when accomplished, but not required for the phenomenological framework to be scientifically useful.

The analogy to Ginzburg-Landau is **[ANALOGY]** . The mathematical content of the consciousness functional is **[HYPOTHESIS]** . The thermodynamic constraints from ETI axioms are **[ESTABLISHED]** .

C Candidate Simulation Protocol

The following simulation program would provide the most direct test of the Level-2 threshold hypothesis on NISQ devices:

1. Prepare an n-qubit cluster state on a superconducting quantum processor with $n \in \{10, 20, 50\}$.

2. Evolve under the Lindblad master equation with measured device parameters (T_1, T_2, γ_ϕ).

3. At each time step t, compute:

$$Q_{\text{coh}}(t) = 1 - e^{-t/T_2} \tag{20}$$

$$I_{\text{int}}(t) = S_{\text{vN}}(\rho_A(t)) / \log_2(\dim \mathcal{H}_A) \tag{21}$$

$$B_{\text{stab}}(t) = \frac{\|[H, \rho(t)]\|}{\|[H, \rho(t)]\| + \sum_k \|\mathcal{D}[L_k]\rho(t)\|} \tag{22}$$

   where $\mathcal{D}[L]\rho = L\rho L^\dagger - \frac{1}{2}\{L^\dagger L, \rho\}$.

4. Identify the time window $[t_0, t_1]$ during which Level-2 thresholds are jointly satisfied, if any such window exists.

5. A null result (no such window exists for any $n \leq 50$) constrains the Level-2 threshold constants from below.

This simulation is a direct extension of the OP3 numerical campaign described in the companion program [1], requiring only the addition of the B_{stab} computation to the existing simulation infrastructure.